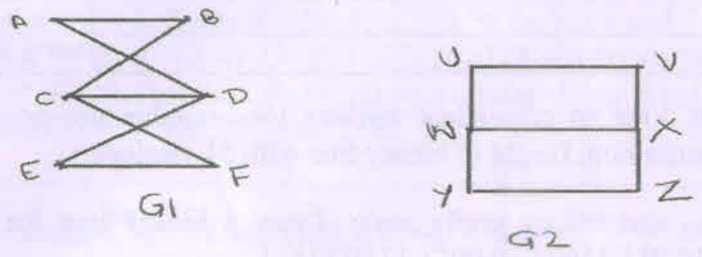
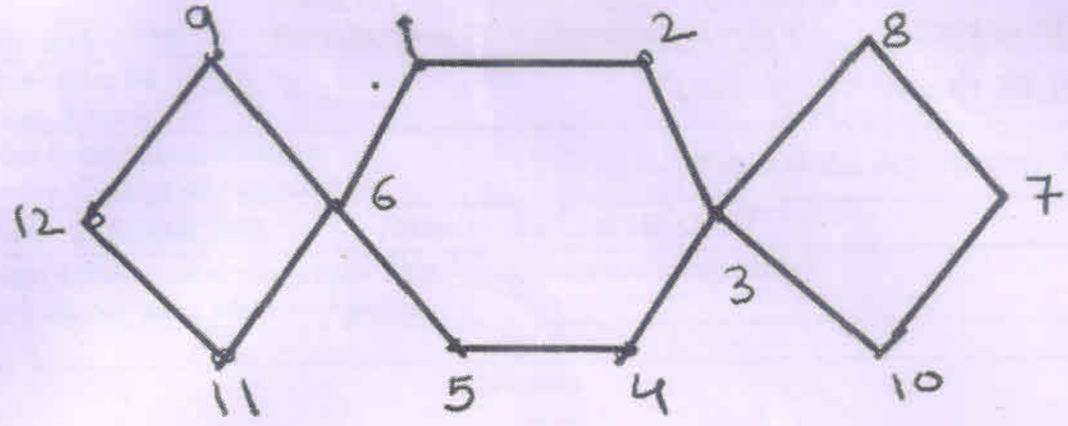
05.06.2023 (E)

Maximum Marks: 100	Semester: January 2023 to May 2023 Examination: ESE Examination – KT		Duration:3 Hrs.
Programme code: 04 Programme: B.Tech.		Class: SY	Semester:_III_(SVU 2020)
Name of the Constituent College: K. J. Somaiya College of Engineering		Name of the department: IT	
Course Code :116U04C301	Name of the Course: DAM		
Instructions: 1)Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary			

Que. No.	Question	Max. Marks
Q1	Solve any **Four**	**20**
i)	Are following graphs isomorphic? Check with justification 	5
ii)	Functions $f: R \to R$, $g: R \to R$ are defined as $f(x) = x - 4$, $g(x) = 6 + 7x$ then find fog , fof , gof & $gogof$	5
iii)	If A= {3,4,5,6} and the relation R defined on the set A as aRb if a+b < 9. (i) Write R as a set of ordered pairs (ii) find matrix of R (iii) Check whether R is reflexive, symmetric, transitive ?	5
iv)	Find Laplace Transform of $\{e^{-3t}(\cos 6t \cos 9t)\}$	5
v)	Obtain Fourier coefficient a_n in the Fourier series of $f(x) = \sqrt{1 - \cos x}$ in $0 < x < 2\pi$ Hence find a_1 & a_2	5
vi)	If 5 points are to be chosen in a square of side 2 units then show that there are at least 2 points at a distant less than $\sqrt{2}$ units	5
Q2 A	Solve the following	**10**
i)	Find Laplace Transform of $(1 + 2t + t^2)H(t-1)$	5
ii)	Find Inverse Laplace Transform of $\tan^{-1}\left(\frac{a}{s}\right)$	5
	OR	

Q2 A	Obtain Complex form of Fourier Series for $f(x) = e^{ax}$ *hence* obtain Complex form of Fourier Series for e^{-ax} & $\sinh(ax)$ $in\ (-l, l)$ where a is not an integer.	10
Q 2 B	Solve any **One**	10
i)	Define Heaviside unit step function. Express the following function as Heaviside unit step function and find [redacted] Laplace transform $$f(t) = \begin{cases} 0, & 0 < t < 1 \\ t+1, & 1 < t < 2 \\ 5t^2, & t > 2 \end{cases}$$	10
ii)	Obtain Fourier Series for $f(x) = \begin{cases} a(x-l), & -l < x < 0 \\ a(x+l), & 0 < x < l \end{cases}$ Hence deduce that $1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} \ldots\ldots\ldots = \frac{\pi}{4}$	10
Q3	Solve any **Two**	20
i)	Find total no of edges, total no of pendant vertices, total number non-pendant of vertices, maximum ,minimum height of binary tree with 51 vertices Define a binary tree and binary prefix code. Draw a binary tree for binary prefixcode {10,000,010,011,110,0010,0011,1110,1111}	10
ii)	Using Laplace transform, solve the following differential equation $$\frac{d^2y}{dt^2} + 4\frac{dy}{dt} + 8y = 1, \quad y(0) = 0, \quad y'(0) = 1$$	10
iii)	If a relation R on the set of integers is defined as aRb if a-b is multiple of 4 Show that the relation is an equivalence relation hence find all equivalence classes	10
Q4	Solve any **Two**	20
i)	State Convolution theorem and use it to find Inverse Laplace Transform of $\left\{\frac{s}{(s^2+4)(s^2+9)}\right\}$	10
ii)	Prepare table with respect to multiplication in S= Z_9-{$\bar{0}$} Determine whether S is a monoid, a semigroup, a group ?	10
iii)	Let A = {1, 2, 3,4}. Let R & S be an equivalence relations on A given as R = {(1, 1), (1, 2), (2, 1), (2, 2), (3,4), (4, 3), (3, 3), (4, 4)} S= {(1, 1), (2, 2), (3,1), (1, 3), (3, 3), (4, 4)} Write $R^{-1} \cap S^{-1}$, S^{-1}, R^{-1} ,SOR find partition of A induced by $R^{-1} \cap S^{-1}$, S^{-1}, R^{-1}, SOR	10
Q5	Solve any **four**	20
i)	Define a Bipartite graph Check given graph is Bipartite graph or not with justification?	5

		
ii)	If f: $R - \{5\} \to R - \{0\}$ is defined as $f(x) = \frac{1}{x-5}$. Show that f(x) is bijective and hence find $f^{-1}(x)$.	5
iii)	Let $A = \{1, 2, 3, 4\}$, $B = \{a, b, c, d\}$, $C = \{x, y, z\}$ and let $R = \{(1, a), (2, d), (3, a), (3, b), (4, a), (3, d)\}$ be a relation from A to B and $S = \{(b, x), (b, z), (c, y), (d, z)\}$ be a relation from B to C. Write SOR Find Domain and Range of SOR	5
iv)	Evaluate $\int_0^{\infty} e^{-3t} t \cos t \, dt$ using Laplace Transform.	5
v)	For half range cosine series of $f(x) = x, 0 \le x \le 2$ If $a_0 = 1$ and $a_n = \frac{4(\cos n\pi - 1)}{n^2 \pi^2}$ then using Parseval's Identity prove that $\frac{\pi^4}{96} = \frac{1}{1^4} + \frac{1}{3^4} + \frac{1}{5^4}$	5
vi)	Define zero divisors and unit elements of Z_n, hence find zero divisors and unit elements of Z_{16}	5

SOMAIYA VIDYAVIHAR UNIVERSITY 09.06.2023 (E)

Maximum Marks: 100	Semester: January 2023 –May 2023 Examination: ESE Examination – KT	Duration:3 Hrs.
Programme code: 04 Programme: BTech IT	Class: SY	Semester: III (SVU 2020)
Name of the Constituent College: K. J. Somaiya College of Engineering	Name of the department: IT	
Course Code: 116U04C303	Name of the Course: Database Management Systems	
Instructions: 1)Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question	Max. Marks
Q1	Solve any **Four**	20
i)	List any 5 advantages of database over file system.	5
ii)	Draw diagram for database system architecture.	5
iii)	Explain the Hierarchical database model in brief.	5
iv)	Define data independence? What is logical data independence?	5
v)	List down all DDL and DML commands.	5
vi)	What can naive users and application programmers do?	5

Que. No.	Question	Max. Marks
Q2 A	Solve the following	10
i)	Explain the concept of generalization with diagram.	5
ii)	What are views in database? Give syntax for create View.	5
	OR	
Q2 A	Explain the domain and referential integrity constraints in database systems.	10
Q 2 B	Solve any **One**	10
i)	Canberra Employment Centre (CEC) places temporary workers in companies during peak periods. CEC maintains a file of candidates who wish to work. If the candidate has worked before, that candidate has a specific job history. (Naturally, no job history exists if the candidate has never worked.)Each candidate may have several qualifications. CEC also has a list of companies that request temporaries. Each time a company requests a temporary employee, CEC makes an entry in the openings folder. This folder contains an opening number, company name, required qualifications, starting date, anticipated ending date, and hourly pay. Each opening requires only one specific qualification Draw an ER diagram that captures the above information, which should include: 1. Identifying the entities, relationships and their attributes 2. indicating the key attributes which you have chosen.	10
ii)	Explain the steps of EER to Relational model conversion with example.	10

Que. No.	Question	Max. Marks
Q3	Solve any **Two**	20
i)	Explain 1NF, 2NF and 3NF with example.	10
ii)	What is indexing? Why it is use in database? Explain with example how indexing cost affects the overall budget of the project.	10
iii)	Explain the steps of query processing and optimization with diagram.	10

Que. No.	Question	Max. Marks
Q4	Solve any **Two**	20
i)	Draw the transaction state-diagram and explain each of its state.	10
ii)	What are ACID properties? Explain with example.	10
iii)	Explain the concept of deadlock and deadlock prevention techniques.	10

Que. No.	Question	Max. Marks
Q5	Write notes on any **four**	20
i)	Conflict serializibility	5
ii)	View serializibilty	5
iii)	Deadlock avoidance techniques.	5
iv)	Two phase locking protocol.	5
v)	Shadow paging.	5
vi)	Validation based protocol.	5

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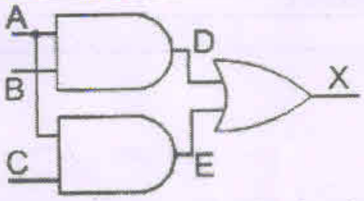
VIDYAVIHAR UNIVERSITY

Maximum Marks: 100	Semester: January 2023 –May 2023 Examination: ESE Examination (KT)	Duration:3 Hrs.
Programme code: 04 Programme: BTech-IT	Class: SY	Semester: III(SVU 2020)
Name of the Constituent College: K. J. Somaiya College of Engineering		Name of the department: Information Technology
Course Code: 116U04C304	Name of the Course: Digital Systems (DIS)	
Instructions: 1)Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question	Max. Marks
Q1	Solve any **Four**	**20**
i)	$(333)_5 = (??)_6$	5
ii)	 Write the Truth Table for the above combinational circuit. Also write the expression for X.	5
iii)	Explain 2-bit Magnitude Comparator	5
iv)	Design 2-bit Synchronous Up counter JK Flip Flop.	5
v)	Explain the concept of Memory Hierarchy with design and characteristics.	5
vi)	Write the name of any 5 addressing modes of 8086 microprocessor with suitable example.	5

Que. No.	Question	Max. Marks
Q2 A	Solve the following	**10**
i)	Design Johnson's counter using D FF.	5
ii)	Design Mod-10 counter using T FF.	5
	asynchronous OR	
Q2 A	Explain 2-bit positive edge trigger Up Counter with timing diagram.	10
Q 2 B	Solve any **One**	**10**
i)	Explain Toggle Flip Flop in detail.	10
ii)	What is Shift Register? Design and explain SISO Shift Register with timing diagram.	10

Que. No.	Question	Max. Marks
Q3	Solve any **Two**	**20**
i)	Implement 4:16 line decoder for following functions a) $F1 = \sum m\ (1, 2, 4, 7, 8, 11, 12, 13)$ b) $F2 = \sum m\ (10, 12, 13, 14)$	10
ii)	Implement 8:1 MUX using only 2:1 MUX.	10
iii)	With any five valid points differentiate between Sequential and Combinational Circuit.	10

Que. No.	Question	Max. Marks
Q4	Solve any **Two**	**20**
i)	Explain the Von Neumann Architecture with suitable diagram.	10
ii)	Explain the concept of Interrupt handling.	10
iii)	Explain the working of DRAM architecture.	10

Que. No.	Question	Max. Marks
Q5	(Write notes / Short question type) on any **four**	**20**
i)	Simplify the Product-Of-Sums Boolean expression below, providing a result in POS form. $Y = (A+B+C+D') . (A+B+C'+D) . (A+B'+C+D') . (A+B'+C'+D)$ $(A'+B'+C'+D) . (A'+B+C+D') . (A'+B+C'+D)$	5
ii)	Discuss the idea of Priority Encoder.	5
iii)	How will you convert SR FF into JK FF?	5
iv)	In brief explain the functional units of computer system.	5
v)	Draw functional block diagram of 8086 microprocessor.	5
vi)	What is segmented memory?	5

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SOMAIYA
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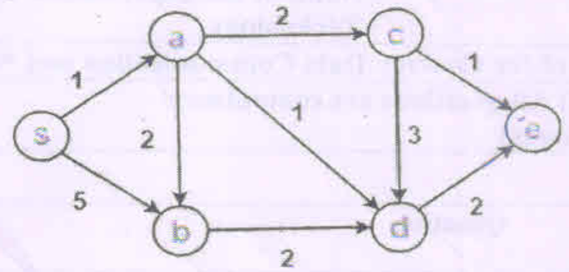
Maximum Marks: 100	Semester: Jan 2023- May 2023 Examination: ESE Examination KT-May23 Duration:3 Hrs.		
Programme code: 04 Programme: B.Tech		Class: S.Y	Semester: III(SVU 2020)
Name of the Constituent College: K. J. Somaiya College of Engineering		Name of the department: Information Technology	
Course Code: 116U04C305	Name of the Course: Data Communication and Networking		
Instructions: 1)Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary			

Que. No.	Question	Max. Marks
Q1	Solve any Four	20
i.	What is protocol? What are its key components?	5
ii.	What is a URL and what are its components? Give an example to interpret the same.	5
iii.	What do you understand by classful addressing? How to find the class address using binary notation?	5
iv.	Describe unicast, broadcast and multicast addressing mechanism use in data Communication networks?	5
v.	What is an error? How does a single-bit error differ from a burst error?	5
vi.	Classify the protocol used over noisy communication channel versus noiseless communication channel?	5
Q2 A	Solve the following	10
i)	What are the differences between TCP and UDP protocol?	5
ii)	What are different routing methods available in data communication and networking?	5
	OR	
Q2 A	What is the purpose of OSI model ? What are the different layers of ISO OSI Model? What are the functions of each layer?	10
Q 2 B	Solve any One	10
i)	What is a HTTP? What are its features? How message or documents transfer from one client to another client using HTTP?	10
ii)	What is client server model? How it works with suitable diagram? State its advantages and disadvantages. .	10

Q3	Solve any Two	20
i)	What is subnet? Why we need subnet? A small organization with given IP block 200.1.2.0/24, needs to creates 2 subnets. Find its subnet mask? Find the number of addresses in each subnet? Find the first address and last address of the block?	10
ii)	Explain the Dijkstra Shortest Path Algorithm.  Find the shortest distance from source vertex 'S' to 'e' vertex using Dijkstra shortest path algorithm?	10
iii)	What is IPV4? Draw IPV4 header diagram. Find out following information for below given IP address: 23.192.157.234 Network Address, , Class of IP Address, Loopback address ,Bits allocated for Network ID and Host ID	10
Q4	Solve any Two	20
i)	Compute transmitted message if the message is 110000101011 using the CRC polynomial x^3+1 to protect it from errors? Also, show checking of code word at the receiver's site.	10
ii)	Compute the checksum at sender and receiver side? Consider the data unit to be transmitted over network is 467F, 726F, 4E49,616E.	10
iii)	State and explain differences between error detection and corrections? Which layer of TCP/IP protocols suits handle error detection and correction mechanism? Explain Working of sliding window protocol.	10
Q5	(Write notes / Short question type) on any four	20
i.	What is a network topology? What are its types? Which is mostly used cabled network topology especially in large businesses?	5
ii.	Which are the devices used in network layer. State its role.	5
iii.	User has to transmit the voice and video over a digital transmission. Assume data is of Analog nature. Which kinds of modulation techniques needed in the given situation? State steps to complete this conversion.	5
iv.	Explain CSMA protocol with suitable diagram.	5
v.	What are differences between guided media and unguided media?	5
vi.	What is a PPP protocol? What are its features?	5

16-6-2023 (E)

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VIDYAVIHAR UNIVERSITY

Maximum Marks: 100	Semester: August 2022 – December 2022 Examination: ESE Examination May 23		Duration:3 Hrs.
Programme code: 46 Programme: Minor in IT		Class: SY	Semester: III (SVU 2020)
Name of the Constituent College: K. J. Somaiya College of Engineering		Name of the department: All	
Course Code: 116m46C401	Name of the Course: Building Blocks of Information Technology		
Instructions: 1)Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary			

Que. No.	Question	Max. Marks
Q1	Solve any **Four**	**20**
i)	Differentiate between Linear Search and Binary Search technique.	5
ii)	What is Algorithm? Discuss in brief Divide and Conquer Algorithms	5
iii)	Describe about the services provided by Data Link Layer.	5
iv)	What is Distributed operating system? Give the description of two types of Distributed OS.	5
v)	What is verification and validation in software testing?	5
vi)	Who are the stakeholders in software architecture	5

Que. No.	Question	Max. Marks
Q2 A	Solve the following	**10**
i)	Explain the working of CDMA technology.	5
ii)	What is Wireless MAN?	5
	OR	
Q2 A	With suitable diagram explain Client Server Architecture.	10
Q 2 B	Solve any **One**	**10**
i)	What is IP addressing? What are the different classes of IP addresses?	10
ii)	What is Local Area Network configuration?	10

Que. No.	Question	Max. Marks
Q3	Solve any **Two**	**20**
i)	What is shell? What are the two major types of shell? What is shell scripting?	10
ii)	Consider the following data with burst time given in milliseconds: Draw Gantt charts for the execution of these processes using FCFS. Find out turnaround time and average waiting time. process / Arrival Time / Burst time p1 / 3 / 4 p2 / 5 / 3 p3 / 0 / 2 p4 / 5 / 1 p5 / 4 / 3	10
iii)	Consider the reference stream 1,2,3,4,2,1,5,6,2,1,2,3,7,6,3,2,1,2,3,6. How many page faults while using FIFO and LRU using 3 frames?	10

Que. No.	Question	Max. Marks
Q4	Solve any Two	20
i)	Explain software design architecture design process.	10
ii)	Explain in brief any two structural patterns.	10
iii)	Explain the concept of architectural style with following points – 1. Definition 2. Basic properties 3. Benefits of using style	10

Que. No.	Question	Max. Marks
Q5	(Write notes / Short question type) on any **four**	**20**
i)	How does the partitioning works for Quick Sort Algorithm?	5
ii)	Give the difference between TCP and UDP.	5
iii)	Describe any two services provided by operating system.	5
iv)	What is Black Box Testing? What are the different types of it?	5
v)	Draw Use Case diagram for movie ticket booking system.	5
vi)	Discuss following in brief (any 1) 1. Iterator Pattern 2. Command Pattern	5

16.6.2023(E)

SOMAIYA
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Maximum Marks: 100	Semester: August 2022 – December 2022 Examination: ESE Examination	Duration:3 Hrs.
Programme code: 66 Programme: Honours program in AI	Class: SY	Semester: III(SVU 2020)
Name of the Constituent College: K. J. Somaiya College of Engineering		Name of the department: IT
Course Code: 116h66C301	Name of the Course: Fundamentals of Data Science	
Instructions: 1)Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question	Max. Marks
Q1	Solve any **Four**	**20**
i)	What is data science and its benefits?	5
ii)	What are the different types of attributes of data?	5
iii)	What are the quality issues associated with data?	5
iv)	What are the difference between classical data analysis and exploratory data analysis?	5
v)	Compute Euclidean distance between the two points where point are given as - P(3,2) and Q(4,1)	5
vi)	What is a Histogram? The table shows life expectancy of 400 people in one village. Draw Histogram for the following data. Age: 30-40, 40-50, 50-60, 60-70, 70-80, 80-90, 90-100 Death: 14, 56, 60, 86, 74, 62, 48	5
Q2 A	Solve the following	**10**
i)	What is the cumulative frequency? Let assume twenty students were asked to note down there working hours per day. Their response in hour mentioned as – 5,6,3,3,4,7,5,2,3,5,2,3,5,6,5,4,4,3,5 and 3 respectively. Compute the relative frequency and the cumulative relative frequency of given data.	5
ii)	What are the differences between Covariance and Correlation of data?	5
	OR	
Q2 A	How to predict missing value using linear regression? Illustrate with suitable example.	10
Q 2 B	Solve any **One**	**10**
i)	What is skewness of data? What is its importance? Explain computation of Skewness with suitable example.	10
ii)	Find the mean, mode, median, first quartile and third quartile of data on given age values of a person? Data series are 13,15,16,16,19,20,20,21,22,22,25,25,25,25,30,33,35,35,35,35,36,40,45,46, 52,70 Calculate the standard deviation and variance of the above data.	10
Q3	Solve any **Two**	**20**
i)	What are the different numerical measures of similarity and dissimilarity of data science? Enlist and explain at least one similarity measurements (metrics) with its formula for below mentioned data, 1)Similarity between data points	10

	2)Similarity between Strings 3)Similarity between two probable distribution 4)Similarity between two sets	
ii)	What is cosine similarity? How to calculate the cosine similarity? Compute the cosine similarity for the two documents given below? Consider 2 keywords as Bitcoin and Money Document A: " Bitcoin Bitcoin Bitcoin Money" Document B: " Money Money Bitcoin Bitcoin"	10
iii)	What is Jaro Similarity? Discuss the working of Jaro with suitable example.	10
Q4	Solve any **Two**	**20**
i)	What are the ways to reduction of data? Explain any one Attribute subset selection method?	10
ii)	What is data discretization? How do you achieve it using clustering technique? Discretize the given data sample A1(2,10),A2(2,5),A3(8,4),A4(5,8),A5(7,5),A6(6,4),A7(1,2),A8(4,9) using K-means clustering algorithm, consider manhattan distance. Assume initial cluster centres are A1 (2,10), A4 (5,8), A7(1,2). Compute the Centre of three clusters after the first iteration only.	10
iii)	What is normalization of data? Normalize the group of data given below using techniques: 245,367,457,516,678 A. Min –Max normalization method (min=0 and max=1) B. Z score normalization C. Decimal Scaling normalization	10
Q5	(Write notes / Short question type) on any **four**	**20**
i)	What do you understand by word visualization? Why we need it in data science?	5
ii)	What are the difference between graph and chart?	5
iii)	Draw pie chart for a class of 200 student where their interest shown in following table:	5
iv)	Write short note on icon based visualization technique.	5
v)	Use line chart to analyse the performance of students over six tests :	5
vi)	What are the differences between scoreboard and dashboard?	5

Table for Q5 iii):

Sports	Cricket	Football	Badminton	Hockey	Other
Total Student	34	50	24	10	82

Table for Q5 v):

RAM	70	89	93	88	85	90
SHAM	72	78	83	90	92	94